AMT Worksheet 11
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The purpose of this assignment is to give you some practice with using the Boltzmann factor and the expression for entropy in terms of the number of atoms occupying a given number of energy states with given energies.

1. As we saw in class, the population of atoms occupying a state of energy ε in thermal equilibrium at temperature T is proportional to $e^{-\beta\varepsilon}$ where $\beta = 1/k_B T$. This can be interpreted as indicating that $k_B T$ represents the amount of energy that is *easily* available to atoms in thermal equilibrium at absolute temperature T. Please use the *electron-volt* as the energy unit throughout this problem. The Boltzmann constant is given by
$$k_B = 8.61733 \times 10^{-5}\ \text{eV/K}.$$
   (a) The bond energy of a hydrogen molecule is given by about 4.52 eV. Determine the Boltzmann factor for this energy at a temperature of 300 K (about 'room temperature'). This can be thought of as a relative probability that hydrogen molecules will dissociate at this temperature.

   (b) A hydrogen molecule requires at least 0.015157 eV of energy in order to rotate about an axis perpendicular to its bond axis. Determine the Boltzmann factor associated with this energy at
   (i) 300 K.
   (ii) 77 K.
   (iii) 4 K.

   (c) A nitrogen molecule requires at least 0.0004933 eV of energy in order to rotate about an axis perpendicular to its bond axis. Determine the Boltzmann factor associated with this energy at
   (i) 300 K.
   (ii) 77 K.
   (iii) 4 K.

   (d) Explain the difference between your answers to parts (b) and (c) in terms of the dependence the heat capacity at constant volume has on temperature.

   (e) Determine the amount of energy, in electron-volts, that is *freely* available (so… this is just the energy $k_B T$) at the following temperatures. Please give your answers as approximate fractions like 1/3 eV.
   (i) 4 K.
   (ii) 77 K.
   (iii) 300 K.
   (iv) 5800 K.

2. Suppose that we have 5×10^{20} atoms, each of which are free to occupy any one of three accessible energy states with energies of $0, \varepsilon$, and 5ε. You may assume that the number of atoms in each of these states throughout this problem is large enough that you can use the Stirling approximation with impunity. All numerical answers should be given in scientific notation correct to three decimal places.
   (a) What configuration will give the largest entropy if we do not have to restrict the total energy of the system? What is the entropy of this configuration?

   (b) Determine the temperature of the system if the total energy is fixed at $2\times10^{20}\varepsilon$. Give your result as a decimal multiple of ε/k_B. How many atoms will be in each state under this restriction, and what will be the resulting entropy?

   (c) Determine the temperature of the system if the total energy is fixed at $7\times10^{20}\varepsilon$. Give your result as a decimal multiple of ε/k_B. How many atoms will be in each state under this restriction, and what will be the resulting entropy?

   (d) Determine the temperature of the system if the total energy is fixed at $20\times10^{20}\varepsilon$. Give your result as a decimal multiple of ε/k_B. How many atoms will be in each state under this restriction, and what will be the resulting entropy? What's different about this situation?

   (e) Suppose that the temperature is fixed at $7\varepsilon/k_B$. How many atoms are in each state? What is the total energy of the system? What is the resulting entropy?

3. When considering gases on a larger scale, we need to include the gravitational force. The potential energy associated with a molecule of mass *m* at height *h* above the surface of the Earth can be taken as $U_g = mgh$, where g = 9.8 meters per square second. In the following, we will ignore the contributions pressure makes to the potential energy of a gas molecule and the fact that the atmosphere is *not* at constant temperature.
   (a) Write a normalized distribution for the probability of finding a gas molecule of mass *m* at height *h* above the surface of the Earth, taking the atmosphere in thermal equilibrium at temperature *T*.

   (b) Determine the average height of a gas molecule of mass *m* in equilibrium at Kelvin temperature *T*. Find the average height of a chlorine, argon, and oxygen molecule/atom in thermal equilibrium at 300 Kelvin, in kilometers, then determine the probability that each of these will be found above 20 kilometers.

4. The Maxwell-Boltzmann distribution is based on the idea that the probability that a state of energy E will be populated in a system in equilibrium at Kelvin temperature T is proportional to $e^{-\beta E}$, where $\beta = 1/k_B T$. The Boltzmann constant $k_B \approx 1.38 \times 10^{-23}\ J/K$ can be thought of as a conversion constant between Kelvin and Joules. One of the simplest applications of this distribution involves the sharing of kinetic energy between gaseous molecules in thermal equilibrium.

   (a) The kinetic energy of a molecule with mass m and speed v is given by $K = \frac{1}{2}mv^2$. Ignoring other contributions to the energy of a system of gaseous molecules with mass m, write an expression proportional to the probability that a molecule will have a velocity lying between (v_x, v_y, v_z) and $(v_x + dv_x, v_y + dv_y, v_z + dv_z)$.

   (b) The probability that a molecule will have *some* velocity pointing in *some* direction is between 0 and ∞ is 1. Use this fact to normalize the probability distribution found in (a). You can either split this integral into three separate integrals or use spherical coordinates to evaluate the integral.

   (c) Use your normalized probability distribution to determine the average speed and the root-mean-square (the square root of the average of the square) speed for molecules of mass m in thermal equilibrium at temperature T. Which of these two is larger? Why aren't they the same?

   (d) Determine the mean speed and root-mean-square speed $\left(v_{rms} \equiv \sqrt{\langle v^2 \rangle}\right)$ of hydrogen, nitrogen, and carbon dioxide molecules in thermal equilibrium in our atmosphere at a temperature of 300 K. For comparison, the escape speed of the Earth is approximately 11,200 m/s.

   (e) Find an integral expression for the probability that a molecule of mass m in thermal equilibrium at temperature T has a speed lying between v and $v + \Delta v$. Do a u-substitution to move all the physical constants (mass, temperature, etc) out front so that their only remnant lies in the integration limits. Use this to determine the probability that the speed of a hydrogen, nitrogen, and carbon dioxide molecule will be found between 400 and 500 meters per second at 300 K. What is the probability that the speed will be *larger* than 600 m/s?